

Lab Session 9

OBJECT

Creating queries, forms and reports in Microsoft Access

INTRODUCTION TO QUERIES

Queries select records from one or more tables in a database so they can be viewed, analyzed, and sorted on a common datasheet. The resulting collection of records, called a dynaset (short for dynamic subset), is saved as a database object and can therefore be easily used in the future. The query will be updated whenever the original tables are updated. Types of queries are *select queries* that extract data from tables based on specified values, *find duplicate* queries that display records with duplicate values for one or more of the specified fields, and *find unmatched* queries display records from one table that do not have corresponding values in a second table.

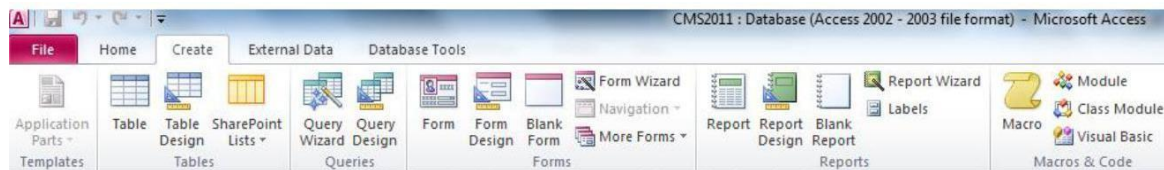


Figure 12.1 Create Menu Quick Access Toolbar

Create a Query in Design View

From the Create Main Menu as in figure 12.1, click the *Query Design* icon. Select tables and existing queries from the *Tables* and *Queries* tabs and click the *Add* button to add each one to the new query. Click *Close* when all of the tables and queries have been selected. Add fields from the tables to the new query by double-clicking the field name in the table boxes or selecting the field from the *Field:* and *Table:* drop-down menus on the query form. Specify sort orders if necessary.

Enter the criteria for the query in the *Criteria:* field. The *Expression Builder*  can also be used to assist in writing the expressions. After you have selected all of the fields and tables, click the *Run* button on the toolbar. Save the query by clicking the *Save* button.

Query Wizard

Click the *Create query by using wizard* icon in the database window to have Access step you through the process of creating a query. From the first window, select fields that will be included in the query by first selecting the table from the drop-down *Tables/Queries* menu. Select the fields by clicking the > button to move the field from the Available Fields list to Selected Fields. Click the double arrow button >> to move all of the fields to Selected Fields. Select another table or query to choose from more fields and repeat the process of moving them to the Selected Fields box. Click *Next >* when all of the fields have been selected. On the next window, enter the name for the query and click *Finish*.

Find Duplicates Query

This query will filter out records in a single table that contain duplicate values in a field. Click the *New* button on the Queries database window, select *Find Duplicates Query Wizard* from the *New Query* window and click *OK*. Select the table or query that the find duplicates query will be applied to from the list provided and click *Next >*. Select the fields that may contain duplicate values by highlighting the names in the Available fields list and clicking the *>* button to individually move the fields to the Duplicate-value fields list or *>>* to move all of the fields. Click *Next >* when all fields have been selected. Select the fields that should appear in the new query along with the fields selected on the previous screen and click *Next >*. Name the new query and click *Finish*.

Delete a Query

To delete a query, select the required query and press the *Delete* key on the keyboard.

FORMS

Forms are used as an alternative way to enter data into a database table.

Create Form by Using Wizard

From the Create Main Menu as in figure 12.1, click *Form wizard* option in Forms section. From the *Tables/Queries* drop-down menu, select the table or query whose datasheet the form will modify. Then, select the fields that will be included on the form by highlighting each one the *Available Fields* window and clicking the single right arrow button *>* to move the field to the *Selected Fields* window. To move all of the fields to *Selected Fields*, click the double right arrow button *>>*. If you make a mistake and would like to remove a field or all of the fields from the *Selected Fields* window, click the left arrow *<* or left double arrow

<< buttons. After the proper fields have been selected, click the *Next >* button to move on to the next screen. On the second screen, select the layout of the form.

- **Columnar** - A single record is displayed at one time with labels and form fields listed side-by-side in columns.
- **Justified** - A single record is displayed with labels and form fields are listed across the screen.
- **Tabular** - Multiple records are listed on the page at a time with fields in columns and records in rows.
- **Datasheet** - Multiple records are displayed in Datasheet View

Click the *Next >* button to move on to the next screen. Select a visual style for the form from the next set of options and click *Next >*. On the final screen, name the form in the space provided. Select "Open the form to view or enter information" to open the form in Form View or "Modify the form's design" to open it in Design View. Click *Finish* to create the form.

Adding Records Using A Form

Input data into the table by filling out the fields of the form. Press the *Tab* key to move from field to field and create a new record by clicking *Tab* after the last field of the last record. A new record can also be created at any time by clicking the *New Record* button  at the bottom of the form window. Records are automatically saved as they are entered so no additional manual saving needs to be executed.

Editing Forms

The following points may be helpful when modifying forms in Design View:

Grid lines - By default, a series of lines and dots underlay the form in Design View so form elements can be easily aligned. To toggle this feature on and off right click and select *Grid*.

Resizing Objects - Form objects can be resized by clicking and dragging the handles on the edges and corners of the element with the mouse.

Change form object type - To easily change the type of form object without having to create a new one, right click on the object with the mouse and select *Change To* and select an available object type from the list.

Label/object alignment - Each form object and its corresponding label are bounded and will move together when either one is moved with the mouse. However, to change the position of the object and label in relation to each other (to move the label closer to a text box, for example), click and drag the large handle at the top, left corner of the object or label.

Tab order - Alter the tab order of the objects on the form by selecting *Tab Order* from design Menu when Form is active. Click the gray box before the row you would like to change in the tab order, drag it to a new location, and release the mouse button.

Form Appearance - Change the background color of the form by clicking the *Shape Fill* button on the format menu inside Form Design. Change the color of individual form objects by highlighting one and selecting a color from

Color palette on the formatting toolbar. The font and size, font effect, font alignment, border around each object, the border width, and a special effect can also be modified using the formatting toolbar buttons inside Form Design Tools.

Form Controls

This section explains the uses for other types of form controls including lists, combo boxes, checkboxes, option groups, and command buttons.

- **List and Combo Boxes:** If there are small, finite number of values for a certain field on a form, using combo or list boxes may be a quicker and easier way of entering data. These two control types differ in the number of values they display. List values are all displayed while the combo box values are not displayed until the arrow button is clicked to open.
- **Check Boxes and Option Buttons:** Use check boxes and option buttons to display yes/no, true/false, or on/off values. Only one value from a group of option buttons can be selected while any or all values from a check box group can be chosen. Typically, these controls should be used when five or fewer options are available. Combo boxes or lists should be used for long lists of options.
- **Command Buttons:** Command buttons provide you with a way of performing action(s) by simply clicking them. When you choose the button, it not only carries out the appropriate action, it also looks as if it's being pushed in and released.

Reports

Reports will organize and group the information in a table or query and provide a way to print the data in a database.

Using the Wizard

From the Create Main Menu as in figure 12.1, click Report Wizard option in the Reports section. Select the information source for the report by selecting a table or query from the *Tables/Queries* drop-down menu. Then, select the fields that should be displayed in the report by transferring them from the *Available Fields* menu to the *Selected Fields* window using the single right arrow button > to move fields one at a time or the double arrow button >> to move all of the fields at once. Click the *Next >* button to move to the next screen.

Select fields from the list that the records should be grouped by and click the right arrow button > to add those fields to the diagram. Use the *Priority* buttons to change the order of the grouped fields if more than one field is selected. Click *Next >* to continue. If the records should be sorted, identify a sort order here. Select the first field that records should be sorted by and click the A-Z sort button to choose from ascending or descending order. Click *Next >* to continue. Select a layout and page orientation for the report and click *Next*

Select a color and graphics style for the report and click *Next >*. On the final screen, name the report and select to open it in either Print Preview or Design View mode. Click the *Finish* button to create the report.

Create in Design View

Click Report Design button in the Reports section of create access toolbar and you will be presented with a blank grid. Automatically properties window will open for this newly being created report. Click on Record Source property combo box and you will be presented with the list of available tables and queries. Choose the data source of the report from the drop-down menu and click OK. Design the report in much the same way you would create a form. For example, double-click the title bar of the Field Box to add all of the fields to the report at once. Then, use the handles on the elements to resize or move them to different locations, and modify the look of the report as per your requirements.

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LAB TASK:

4. Create the following tables in a Database named *Job Details* and add at least 5 records.
➤ *Employees(employeeID, employeeName, hiringDate, deptID, jobID)*

employee ID	employee Name	hiring date	dept ID	job ID	Click to Add
123	Fahad	27/11/2022	6	20	
145	Lisa	08/06/2022	3	77	
456	Hareem	22/08/2008	1	3	
789	Isha	03/05/2005	4	56	
980	Haley	13/01/2023	4	67	
*					

➤ *Departments(deptID, deptName, locationID)*

Dept ID	Dept Name	Location ID	Click to Add
1	HR	g3	
2	Finance	f2	
3	Marketing	h4	
4	Sales	e1	
*			

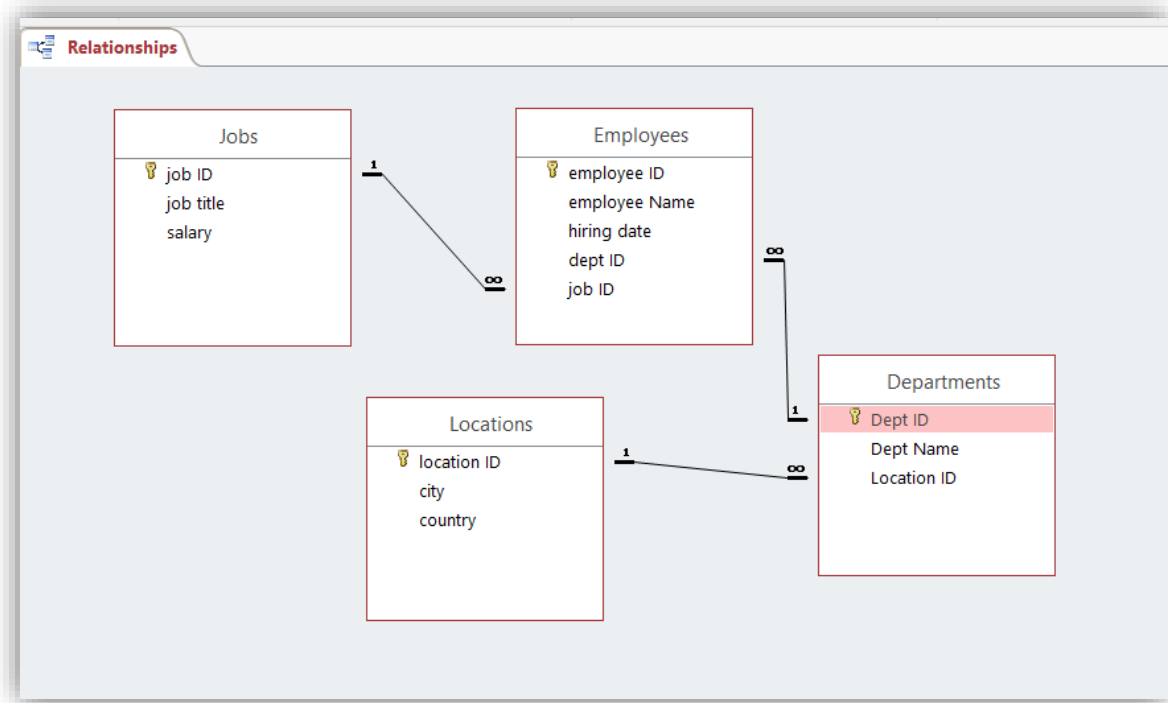
➤ *Jobs(jobID, jobTitle, salary)*

job ID	job title	salary	Click to Add
1	Manager	Rs20,000.00	
2	Analyst	Rs15,000.00	
3	Developer	Rs10,000.00	
4	Technician	Rs5,000.00	
*		Rs0.00	

➤ *Locations(locationID, city, country)*

location ID	city	country	Click to Add
e1	Karachi	Pakistan	
f2	Lahore	Pakistan	
g3	New York	USA	
h4	Toronto	Canada	
*			

5. Create appropriate relations between the newly created tables. Ensure referential integrity.



6. Write suitable queries for each of the following statements.

(i) Display all the records from the Jobs table.

The screenshot shows the Microsoft Access interface with the Query1 window open. The Jobs table is selected in the table list. The field list for the Jobs table is displayed, showing job ID, job title, and salary. The field list is as follows:

Field:	Table:	Sort:	Show:	Criteria:	or:
job ID	Jobs		☒		
job title	Jobs		☐		
salary	Jobs		☐		

job ID	job title	salary		
1	Manager	Rs20,000.00		
2	Analyst	Rs15,000.00		
3	Developer	Rs10,000.00		
4	Technician	Rs5,000.00		
*		Rs0.00		

(ii) Display the hiring date of the employees with department ID = "4".

Employees				
employee ID employee Name hiring date dept ID job ID				

Field:	employee Name	hiring date	dept ID	
Table:	Employees	Employees	Employees	
Sort:				
Show:	☒	☒	☒	☐
Criteria:			"4"	
or:				

employee Name	hiring date	dept ID		
Isha	03/05/2005	4		
Haley	13/01/2023	4		
*				

(iii) Display job IDs for which the salary is greater than “\$1000” or Title is “Manager”.

Employees Departments Jobs Locations Query2 Query1 Query3

Jobs

*

job ID

job title

salary

Field:	job ID	salary	job title		
Table:	Jobs	Jobs	Jobs		
Sort:					
Show:	☒	☒	☒	☐	☐
Criteria:		1000			
or:			"Manager"		

	Employees	Departments	Jobs	Locations
	job ID	salary	job title	
1		Rs20,000.00	Manager	
*		Rs0.00		

(iv) Display the location IDs and cities of the departments that operate in Country of Pakistan.

Employees Departments Jobs Locations Query2 Query1 Query3 Query4 Query5 Query6

Departments

*

Dept ID

Dept Name

Location ID

Locations

*

location ID

city

country

Field:	Location ID	city	country			
Table:	Departments	Locations	Locations			
Sort:						
Show:	☒	☒	☒	☐	☐	☐
Criteria:			"Pakistan"			
or:						

Location ID	city	country	
e1	Karachi	Pakistan	
f2	Lahore	Pakistan	
*			

(v) Display the highest paid salary.

Employees Departments Jobs Locations Query2 Query1 Query3 **Query4** Query5 Query6

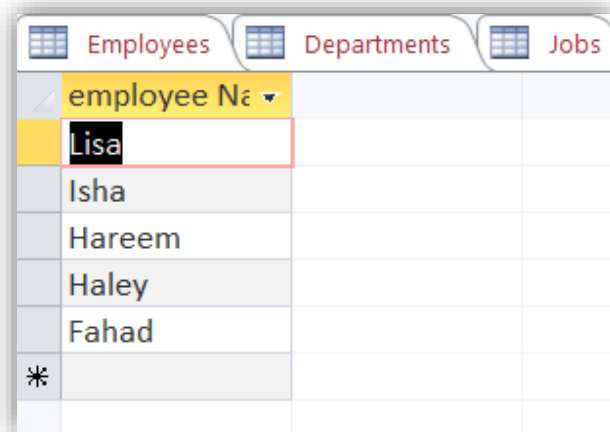
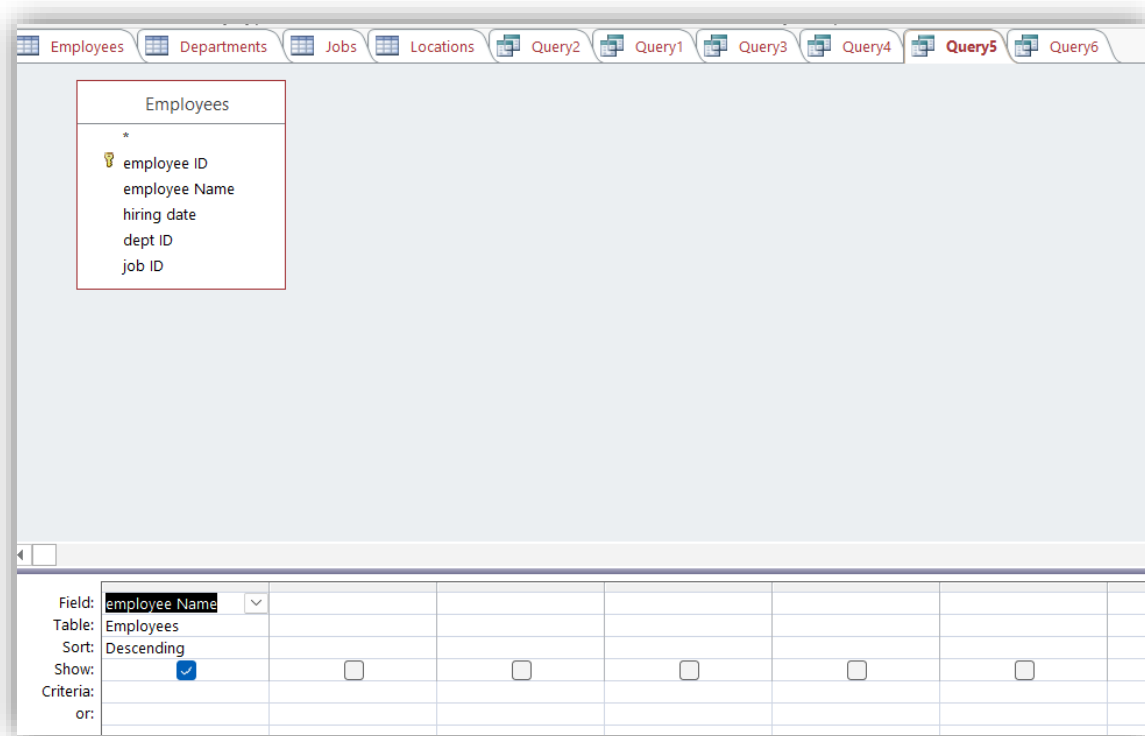
Jobs

job ID
job title
salary

Field: salary							
Table: Jobs							
Total: Max							
Sort:							
Show: ☒	☐	☐	☐	☐	☐	☐	
Criteria:							
or:							

Employees	Departments	Jobs	Locations	Query
MaxOfsalary				
Rs20,000.00				

(vi) Display the names of the employees in reverse alphabetical order.



SUBMISSION GUIDELINES

- Take a screenshot of each task.
- Place all the screenshots in a single word file labeled with Roll No and Lab No. e.g. 'SE202xxx_Lab09'.
- Convert the file into PDF.
- Place all the related files along with the PDF file in a folder labeled with Roll No and Lab No. e.g. 'SE202xxx_Lab09'.
- Submit the folder at [GCR](#)
- -100% policies for plagiarism.

NED University of Engineering & Technology
Department of Software Engineering

Course Code & Title: CT-174 FIT Assessment
Rubric for Psychomotor (P2-PLO-3)

Instructor: _____

Batch: 2024

Class: FESE

Semester: Fall 2024

Student's Name: _____

Roll No.: _____

Laboratory Session No. _____

Date: _____

Psychomotor Domain Assessment Rubric-Level P2-PLO3					
Skill Sets	Extent of Achievement				
	Below Average (0--1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Problem Definition and Understanding	Unable to define and understand the problem clearly.	Struggles to define the problem clearly, with significant gaps in understanding	Defines the problem, but with gaps or a limited understanding of key components, impacting the overall quality of definition.	Effectively defines the problem with a good understanding of its components, showcasing proficiency in problem definition.	Demonstrates an exceptional understanding of the problem, identifying all key components and challenges with insightful depth.
Technical Implementation <i>Proficient use of tools to complete the problem</i>	Struggles to use tools effectively, requiring significant improvement.	Uses tools with noticeable inefficiencies.	Demonstrates basic proficiency in utilizing tool, with some room for improvement.	Effectively uses tools to support the task.	Integrates use of tools seamlessly to enhance the design and development process
Solution Design& Accuracy	Struggles to relate the implemented solutions to the design and development goals, requiring significant improvement.	Shows limited connection between the implemented solutions and the design and development context.	Demonstrates a basic understanding of how the solutions align with design and development requirements.	Effectively relates the implemented solutions to the overall design and development objectives	Maps the implemented solutions seamlessly to the overarching design and development goals, demonstrating a comprehensive understanding.
Trouble Shooting Skills Issue resolution skills	Unable to trouble shoot	Limited troubleshooting	Adequate troubleshooting, minor errors remaining	Good troubleshooting, no errors remaining	Excellent trouble shooting no errors remain without delay.
Document Organization and on time submission	Poor Unclear disorganized document late submission	Somewhat clear with lack of organization and structuring late submission	Somewhat clear with lack of organization and structuring on time submission	Reasonably clear and organized with adequate structuring on time submission	Highly clear and well-structured document, On time submit

Weighted CLO (Psychomotor Score)		
Instructor's Signature with Date		